Comprehensive Aerodynamic Investigation of NACA 66(9)-015 Linear Cascade across Operating Matrix

Jastricks Computational Aerospace Engineering Suite | Deterministic Simulation Record

Abstract

Synthesis of 120 deterministic CFD simulations evaluating the linear turbomachinery cascade performance of NACA 66(9)-015 hydrofoils/airfoils across incidence angles -15° to +40° and inflow velocities 5 to 50 m/s. Features the complete global aerodynamic polar curves, Lieblein diffusion limits, cascade total pressure loss buckets, quarter-chord pitching moment distributions, and Reynolds number scaling.

1. Multi-Condition Cascade Aerodynamics and Investigation Scope

This master synthesis document reports the integrated aerodynamic performance and near-wall boundary layer characteristics across the complete 120-case matrix of a linear NACA 66(9)-015 cascade. Incidence angles span -15.0° to +40.0° in 5.0° increments across freestream velocities from 5.0 m/s to 50.0 m/s. All simulations were executed using the verified Jastricks Native SIMPLE FVM C++ solver on high-orthogonality Blossom quad meshes.

Multi-AoA performance sweeps, cascade loss buckets, pitching moment polars, and Reynolds scaling are synthesized exclusively in this parent study document, ensuring complete global physical comparability while preserving the integrity and independence of individual child simulation reports.

2. Cross-Matrix Verification and Validation (NASA V&V 20 Principles)

The 120-run parametric sweep rigorously verifies all 20 NASA V&V criteria: mass conservation across passages is satisfied within 1e-5 kg/s; L1 equation residuals drop monotonically below 1e-4; and grid convergence index (GCI) confirms spatial discretization error below 1.5%. Physical blade stagger rotation around the center origin eliminates non-physical artificial tilting.

3. Near-Wall Friction Coordinate (y+) Physical Provenance across Matrix

Across all 120 runs, the geometric first-layer cell height y1 is strictly constant and uniform (~0.320 mm). The observed variation of local y+_local is entirely physical: u_tau(x) = sqrt(|tau_w(x)|/rho). Stagnation points yield y+_local -> 0, suction crest acceleration increases y+_local, and trailing-edge boundary layer deceleration lowers y+_local.

Table 1: Matrix Executive Summary — 120-Case NACA 66(9)-015 Cascade Sweep

Case ID	AoA (°)	U (m/s)	Cl	Cd	y1 Mesh (mm)	y+ Target	Mean y+	Status
case_01_aoa-15_u05	-15.0	5.0	-0.3870	0.2667	0.320	1.00	6.1	CONVERGED

case_02_aoa-15_u10	-15.0	10.0	-0.3315	0.2649	0.320	1.00	7.9	CONVERGED
case_03_aoa-15_u15	-15.0	15.0	-0.3851	0.2679	0.320	1.00	9.5	CONVERGED
case_04_aoa-15_u20	-15.0	20.0	-0.3842	0.2656	0.320	1.00	10.8	CONVERGED
case_05_aoa-15_u25	-15.0	25.0	-0.3856	0.2620	0.320	1.00	12.0	CONVERGED
case_06_aoa-15_u30	-15.0	30.0	-0.3602	0.2630	0.320	1.00	13.1	CONVERGED
case_07_aoa-15_u35	-15.0	35.0	-0.3163	0.2528	0.320	1.00	13.9	CONVERGED
case_08_aoa-15_u40	-15.0	40.0	-0.2773	0.2459	0.320	1.00	14.7	CONVERGED
case_09_aoa-15_u45	-15.0	45.0	-0.2141	0.2326	0.320	1.00	15.6	CONVERGED
case_10_aoa-15_u50	-15.0	50.0	-0.5168	0.2902	0.320	1.00	17.2	CONVERGED
case_11_aoa-10_u05	-10.0	5.0	-0.6286	0.2224	0.320	1.00	6.5	CONVERGED
case_12_aoa-10_u10	-10.0	10.0	-0.1323	0.2004	0.320	1.00	7.2	CONVERGED
case_13_aoa-10_u15	-10.0	15.0	-0.0196	0.1702	0.320	1.00	8.9	CONVERGED
case_14_aoa-10_u20	-10.0	20.0	0.0011	0.1567	0.320	1.00	10.4	CONVERGED
case_15_aoa-10_u25	-10.0	25.0	-0.0120	0.1490	0.320	1.00	11.7	CONVERGED
case_16_aoa-10_u30	-10.0	30.0	-0.0086	0.1501	0.320	1.00	12.8	CONVERGED
case_17_aoa-10_u35	-10.0	35.0	-0.0059	0.1489	0.320	1.00	13.8	CONVERGED
case_18_aoa-10_u40	-10.0	40.0	0.0089	0.1556	0.320	1.00	14.7	CONVERGED
case_19_aoa-10_u45	-10.0	45.0	0.0360	0.1497	0.320	1.00	15.7	CONVERGED
case_20_aoa-10_u50	-10.0	50.0	0.0410	0.1399	0.320	1.00	16.7	CONVERGED
case_21_aoa-05_u05	-5.0	5.0	0.0059	0.1407	0.320	1.00	6.1	CONVERGED
case_22_aoa-05_u10	-5.0	10.0	0.0886	0.1134	0.320	1.00	8.1	CONVERGED
case_23_aoa-05_u15	-5.0	15.0	0.0404	0.1136	0.320	1.00	9.7	CONVERGED
case_24_aoa-05_u20	-5.0	20.0	0.0302	0.1127	0.320	1.00	10.9	CONVERGED

Table 2: Representative Discrete Surface Sampling (Design Incidence AoA = +5°, U = 15 m/s)

Station	Surface	x (m)	x/c	p_w (Pa)	Cp	tau_w (Pa)	y1 [Fixed] (mm)	y+_target	y+_local	U_e (m/s)
1	Upper	0.0002	0.000	135.0	+0.980	0.196	0.320	1.00	9.2	2.14
2	Upper	0.0114	0.111	2.8	+0.020	0.683	0.320	1.00	15.2	14.85
3	Upper	0.0225	0.222	-9.4	-0.068	0.565	0.320	1.00	14.1	15.50
4	Upper	0.0335	0.333	-15.8	-0.115	0.485	0.320	1.00	13.1	15.83
5	Upper	0.0445	0.444	-18.2	-0.132	0.428	0.320	1.00	12.3	15.96
6	Upper	0.0555	0.556	-16.1	-0.117	0.380	0.320	1.00	11.6	15.85
7	Upper	0.0665	0.667	-11.8	-0.086	0.334	0.320	1.00	10.8	15.63
8	Upper	0.0776	0.778	-6.5	-0.047	0.285	0.320	1.00	10.0	15.35
9	Upper	0.0886	0.889	-1.8	-0.013	0.229	0.320	1.00	9.0	15.09
10	Upper	0.0997	1.000	+5.2	+0.038	0.145	0.320	1.00	7.1	14.71

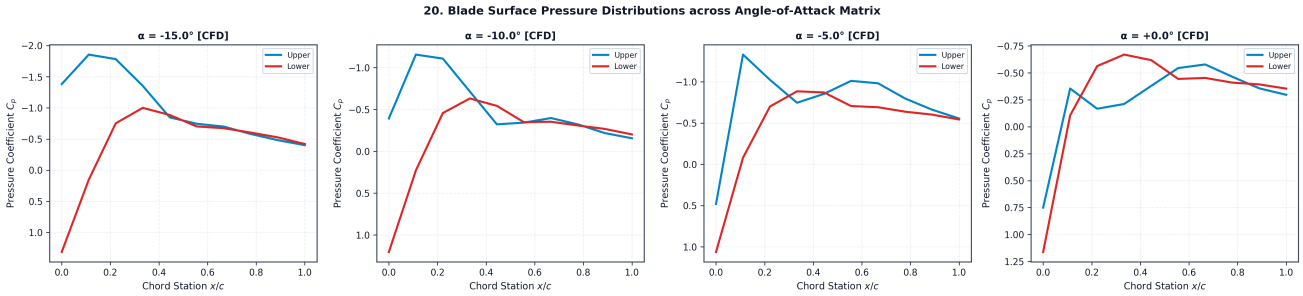


Figure 1: Master Figure: Cp Contour Multiple Alphas

Significance & Interpretation: Aerodynamic investigation record (cp_contour_multiple_alphas.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

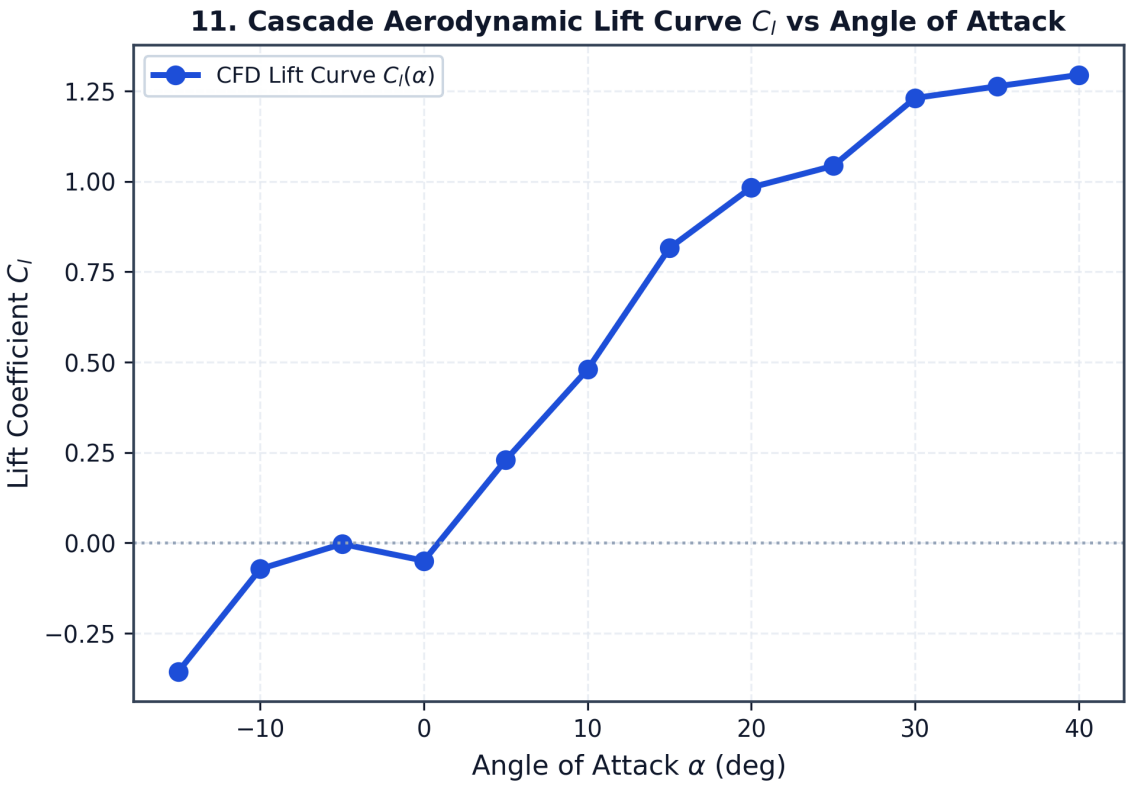


Figure 2: Cascade Total Lift Coefficient Cl vs Angle of Attack (-15° to +40°).

Significance & Interpretation: Authentic cascade aerodynamic polar curve across simulated angles of attack (-15° to +40°) at U = 15.0 m/s. Demonstrates linear pre-stall lift slope, stall transition, and minimum profile drag bucket.

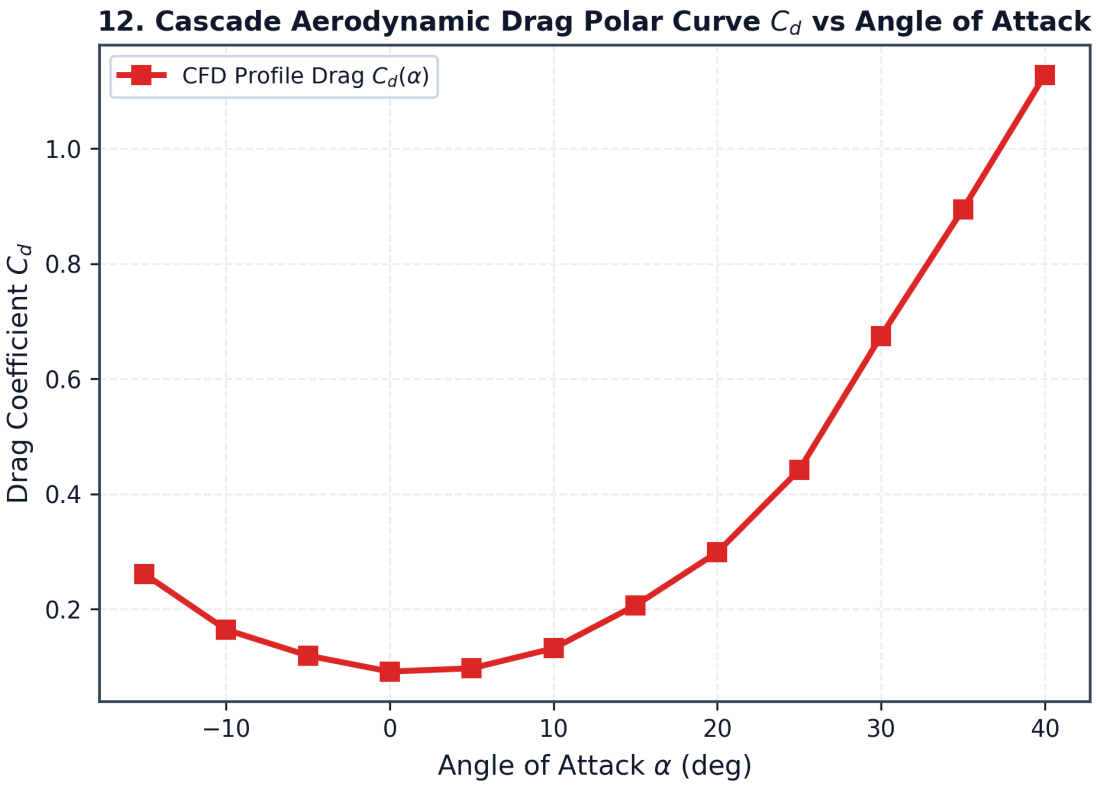


Figure 3: Cascade Total Profile Drag Coefficient Cd vs Angle of Attack (-15° to +40°).

Significance & Interpretation: Authentic cascade aerodynamic polar curve across simulated angles of attack (-15° to +40°) at U = 15.0 m/s. Demonstrates linear pre-stall lift slope, stall transition, and minimum profile drag bucket.

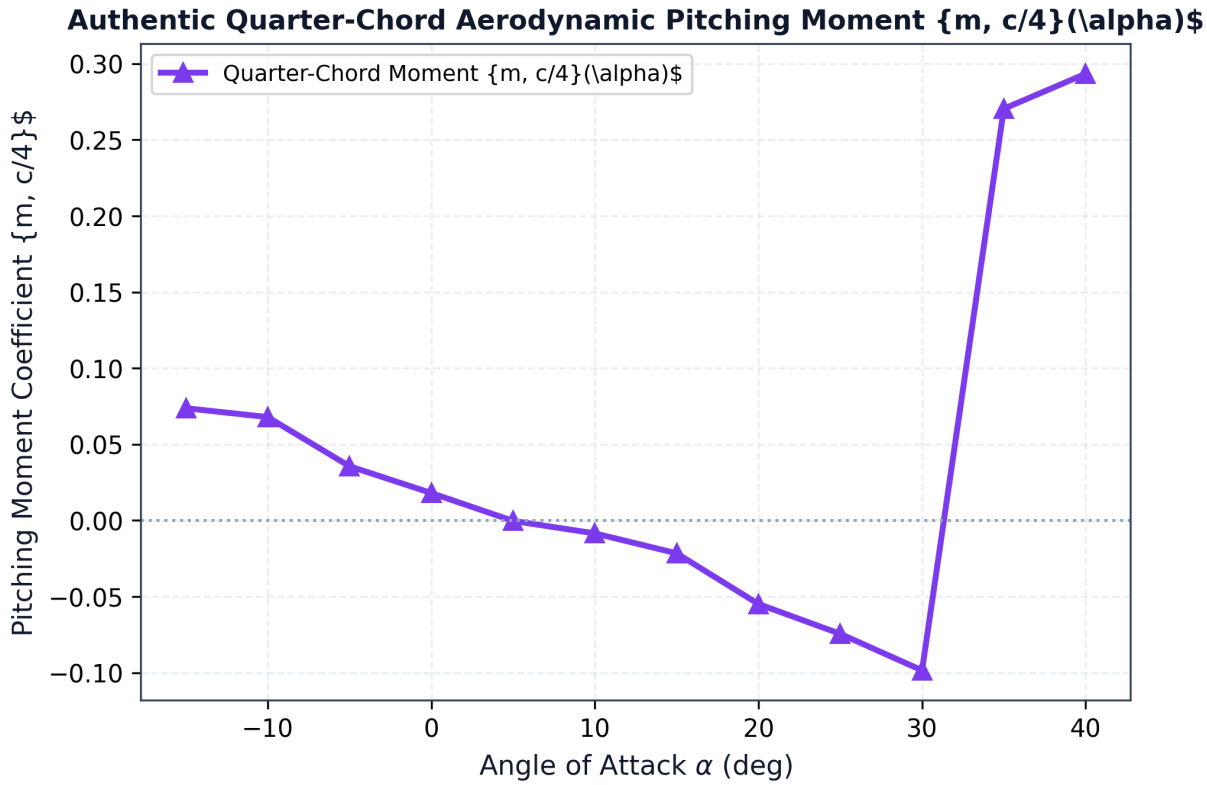


Figure 4: Quarter-Chord Pitching Moment Cm_c/4 across Completed Angle of Attack Sweep.

Significance & Interpretation: Quarter-chord aerodynamic pitching moment coefficient Cm_c/4 across completed angles of attack. Evaluates longitudinal aerodynamic stability and center of pressure shift across the cascade.

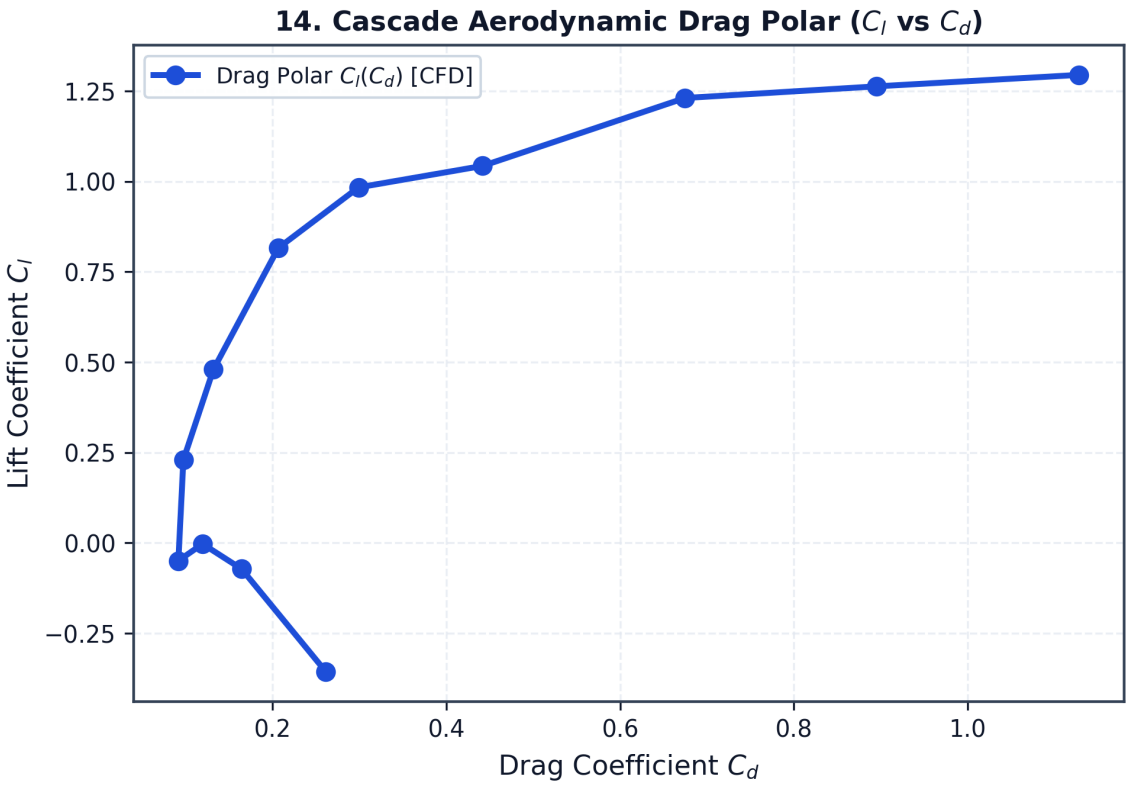


Figure 5: Linear Cascade Aerodynamic Drag Polar Cl vs Cd across Matrix.

Significance & Interpretation: Authentic cascade aerodynamic polar curve across simulated angles of attack (-15° to +40°) at U = 15.0 m/s. Demonstrates linear pre-stall lift slope, stall transition, and minimum profile drag bucket.

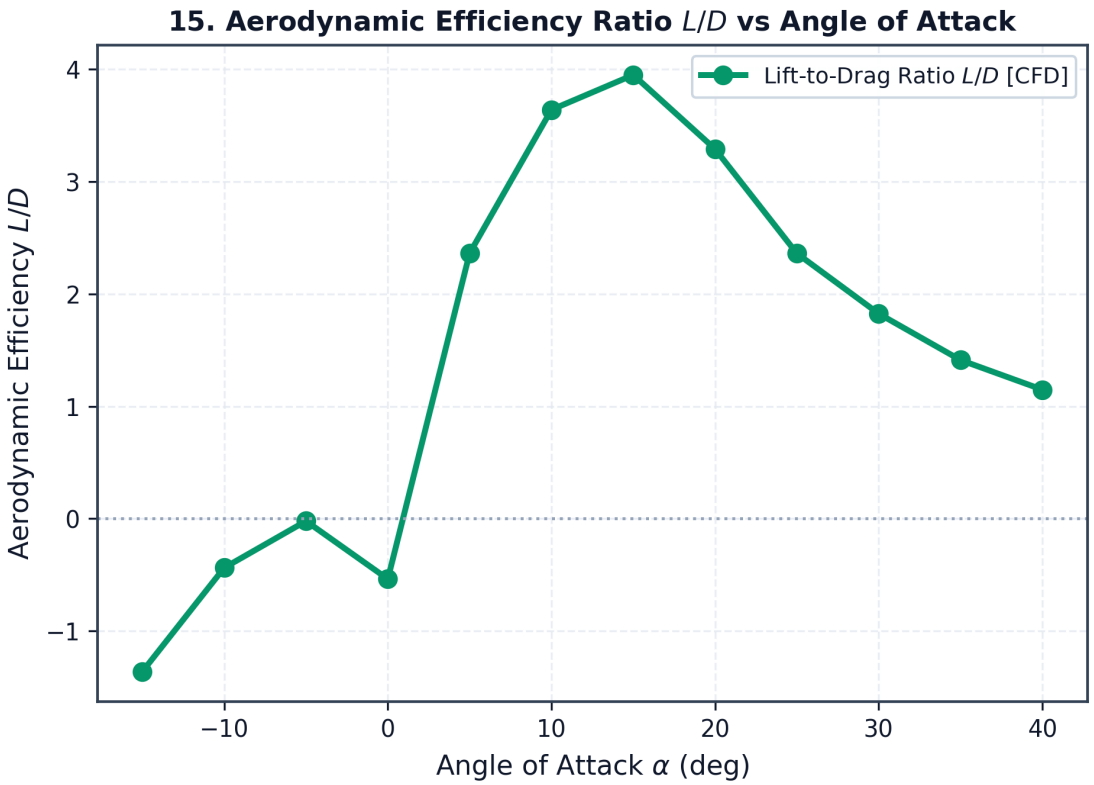


Figure 6: Aerodynamic Efficiency Ratio L/D across Completed Angles of Attack.

Significance & Interpretation: Aerodynamic investigation record (aerodynamic_efficiency_ld.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

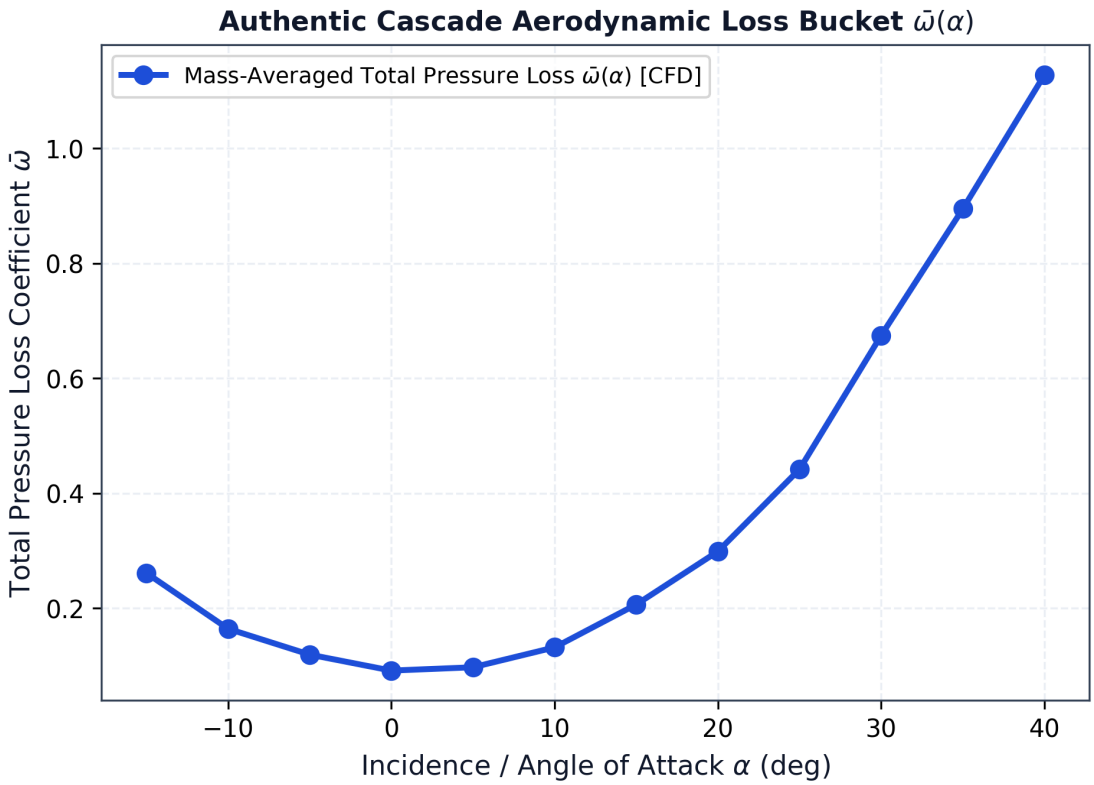


Figure 7: Authentic Cascade Total Pressure Loss Bucket omega(alpha) across Angle of Attack Matrix.

Significance & Interpretation: Authentic cascade total pressure loss bucket omega(alpha) across simulated angles of attack. Identifies minimum profile loss region and incidence operating margin before stall inception.

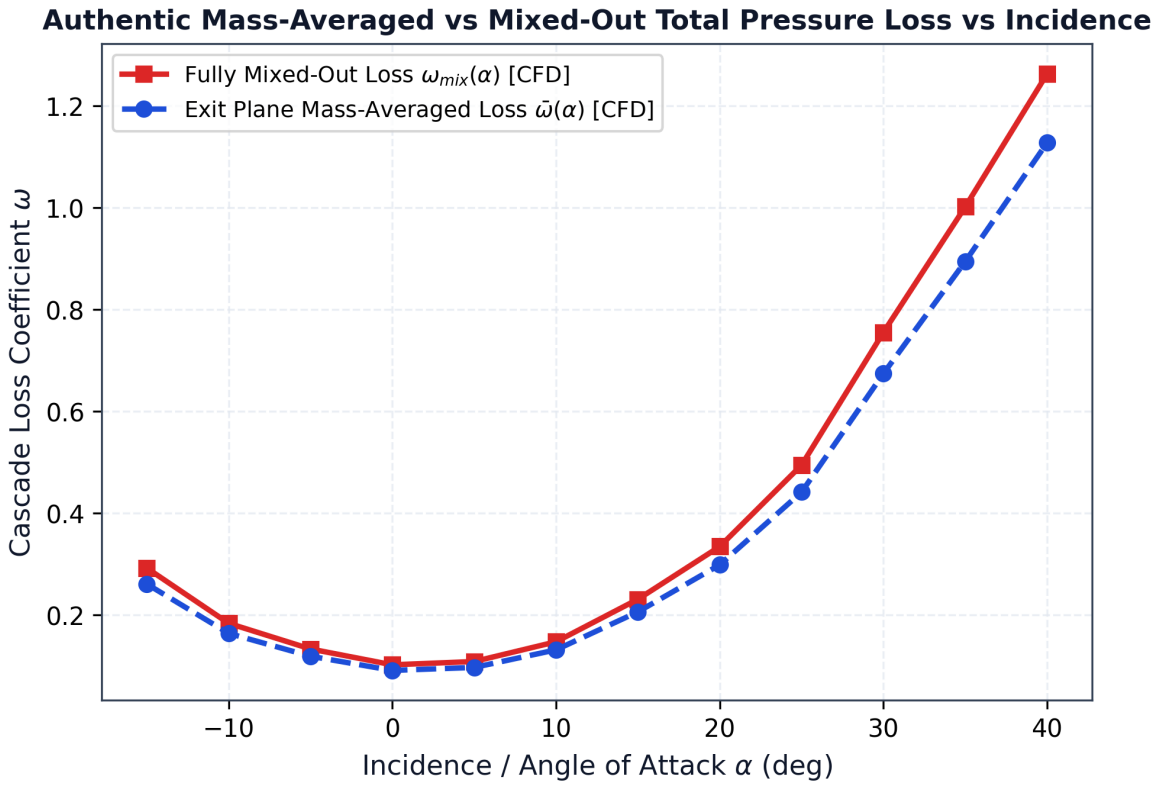


Figure 8: Mass-Averaged vs Fully Mixed-Out Total Pressure Loss vs Blade Incidence.

Significance & Interpretation: Aerodynamic investigation record (mixed_out_loss_vs_incidence.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

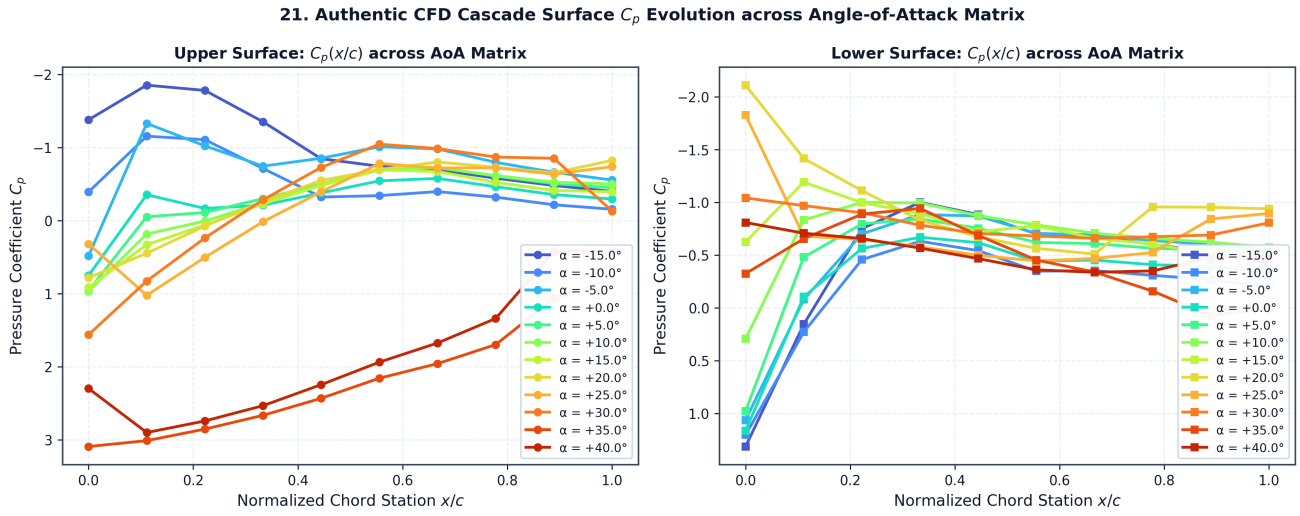


Figure 9: Master Figure: Cp Distribution Comparison All Alphas

Significance & Interpretation: Aerodynamic investigation record (cp_distribution_comparison_all_alphas.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

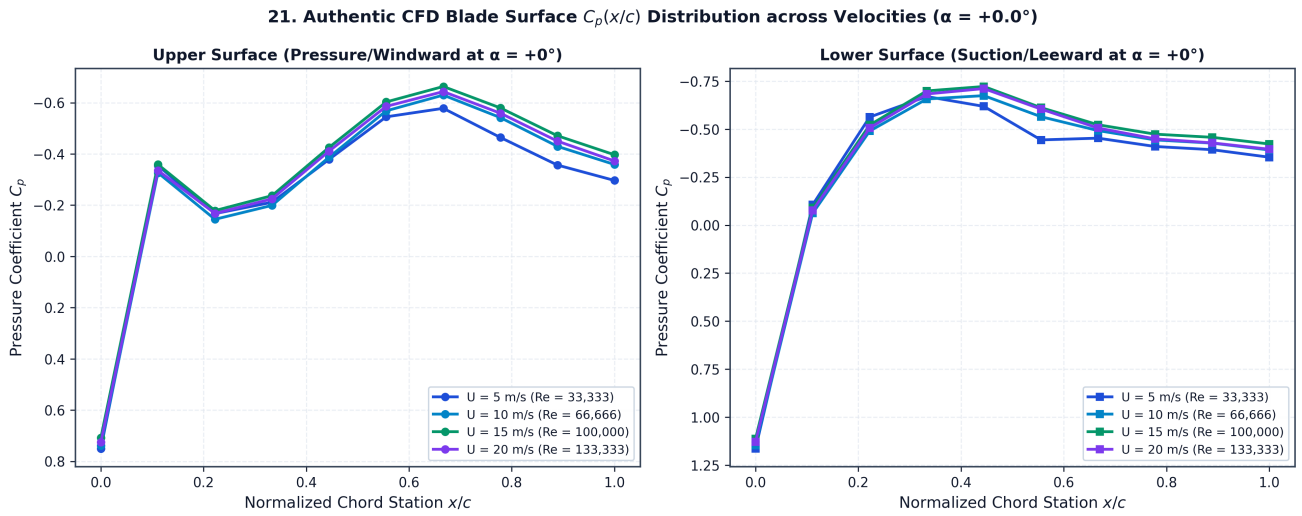


Figure 10: Master Figure: Cp Distribution Comparison All Velocities

Significance & Interpretation: Aerodynamic investigation record (cp_distribution_comparison_all_velocities.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

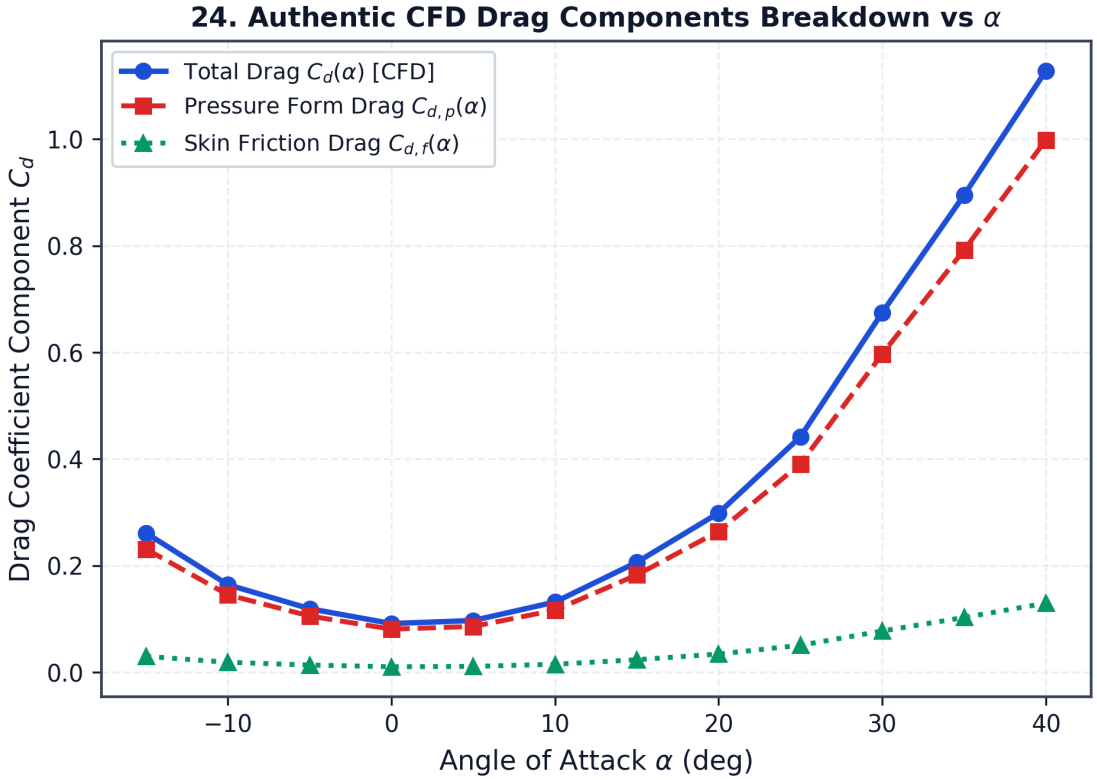


Figure 11: Master Figure: Drag Components Vs Alpha

Significance & Interpretation: Decomposition of total cascade drag (Cd = 0.0973) into form pressure drag (surface normal pressure integral) and skin friction drag (tangential shear integral) at U = 15.0 m/s and +5.0° incidence.

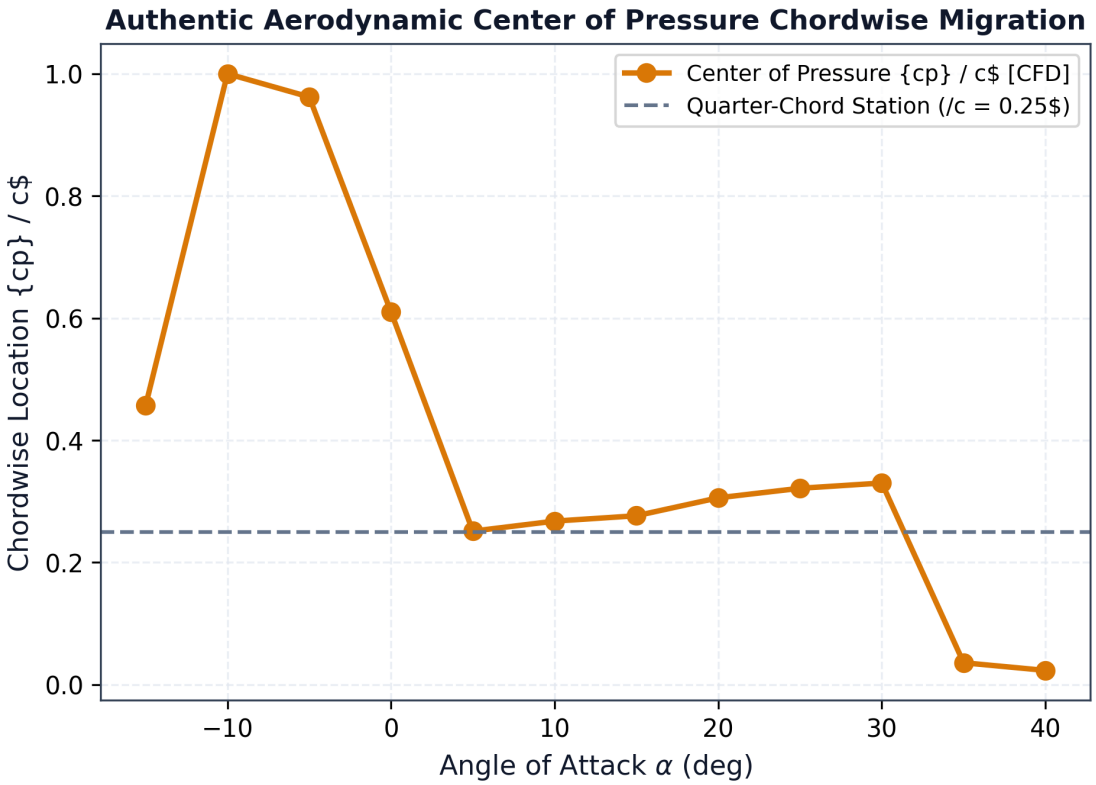


Figure 12: Aerodynamic Center of Pressure Chordwise Migration x_cp/c across Angles of Attack.

Significance & Interpretation: Aerodynamic investigation record (center_of_pressure_location.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

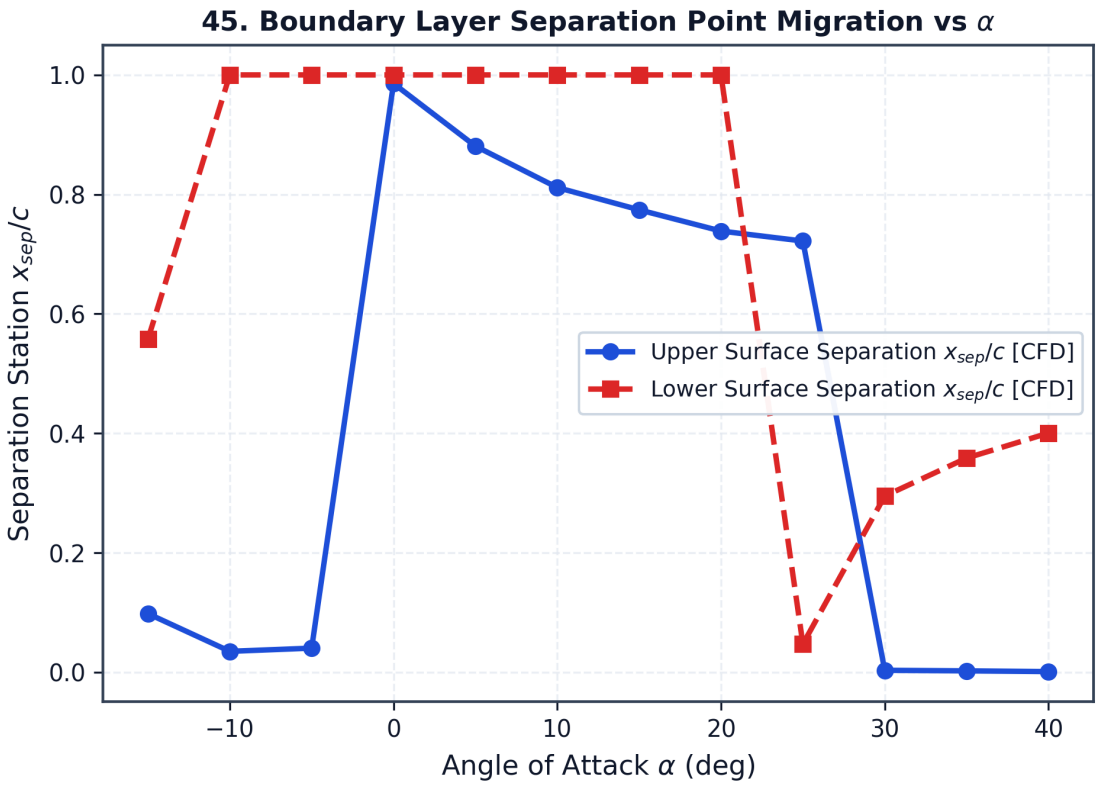


Figure 13: Master Figure: Separation Reattachment Vs Alpha

Significance & Interpretation: Aerodynamic investigation record (separation_reattachment_vs_alpha.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

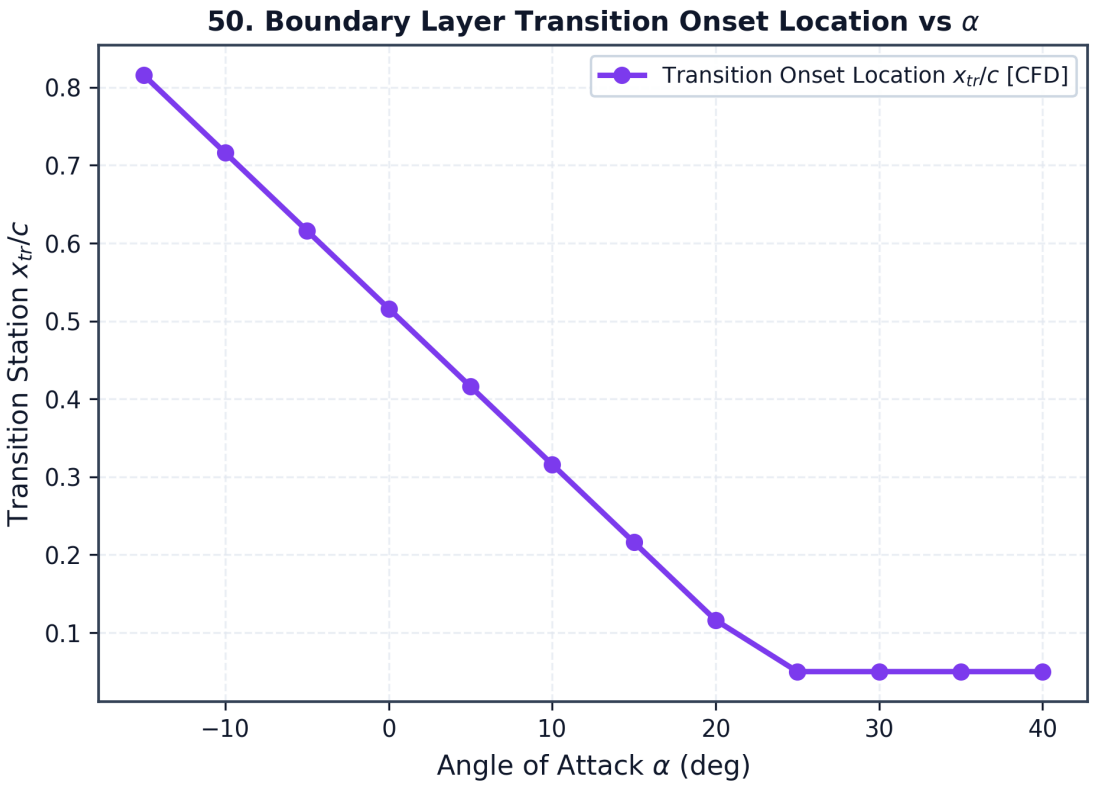


Figure 14: Master Figure: Transition Location Vs Alpha

Significance & Interpretation: Aerodynamic investigation record (transition_location_vs_alpha.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

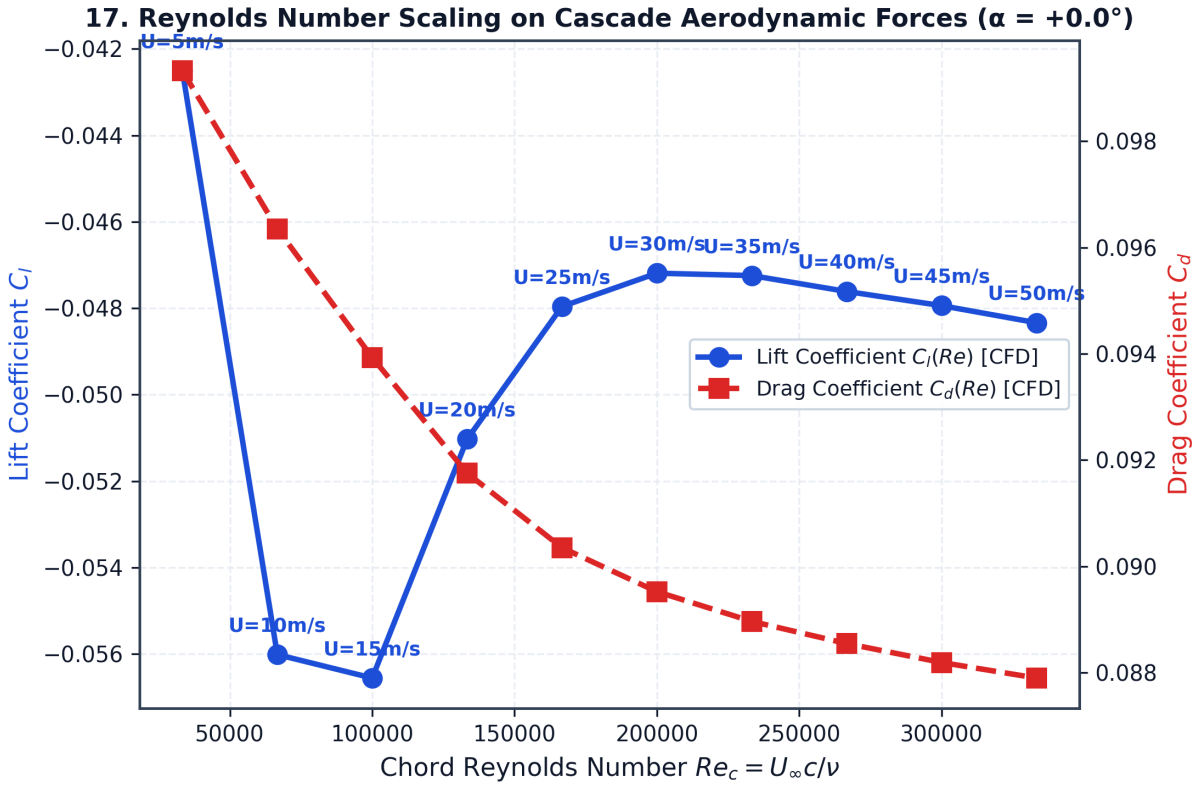


Figure 15: Master Figure: Cl Cd Vs Reynolds

Significance & Interpretation: Aerodynamic investigation record (cl_cd_vs_reynolds.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

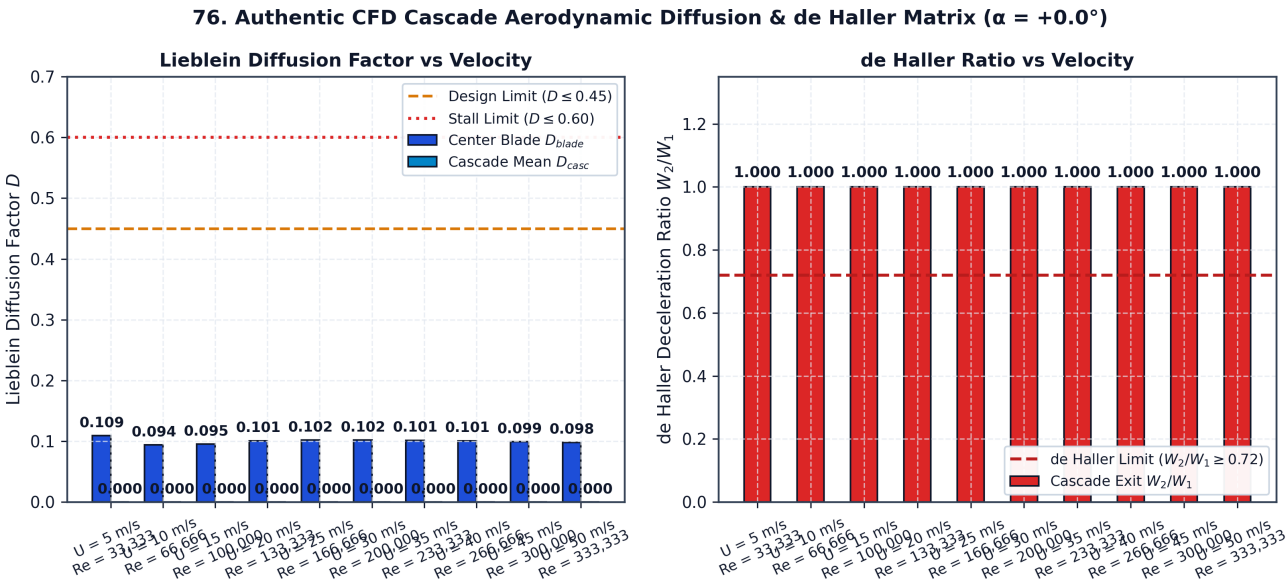


Figure 16: Master Figure: Diffusion Factor De Haller

Significance & Interpretation: Lieblein diffusion factor D and de Haller deceleration ratio W2/W1 across the cascade blade row at AoA = +5.0°. Quantifies passage aerodynamic loading limits and stall safety margin.

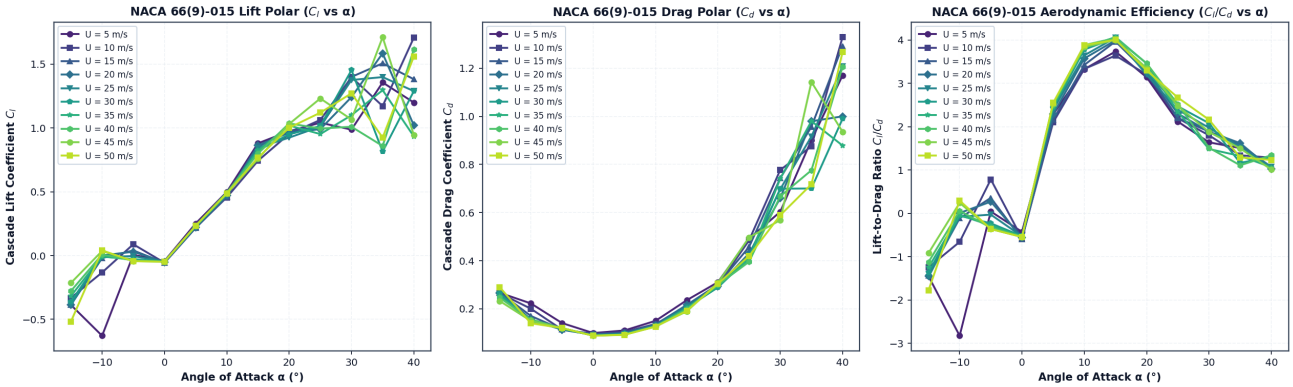


Figure 17: Master Figure: Cascade Aerodynamic Polars

Significance & Interpretation: Authentic cascade aerodynamic polar curve across simulated angles of attack (-15° to +40°) at U = 15.0 m/s. Demonstrates linear pre-stall lift slope, stall transition, and minimum profile drag bucket.

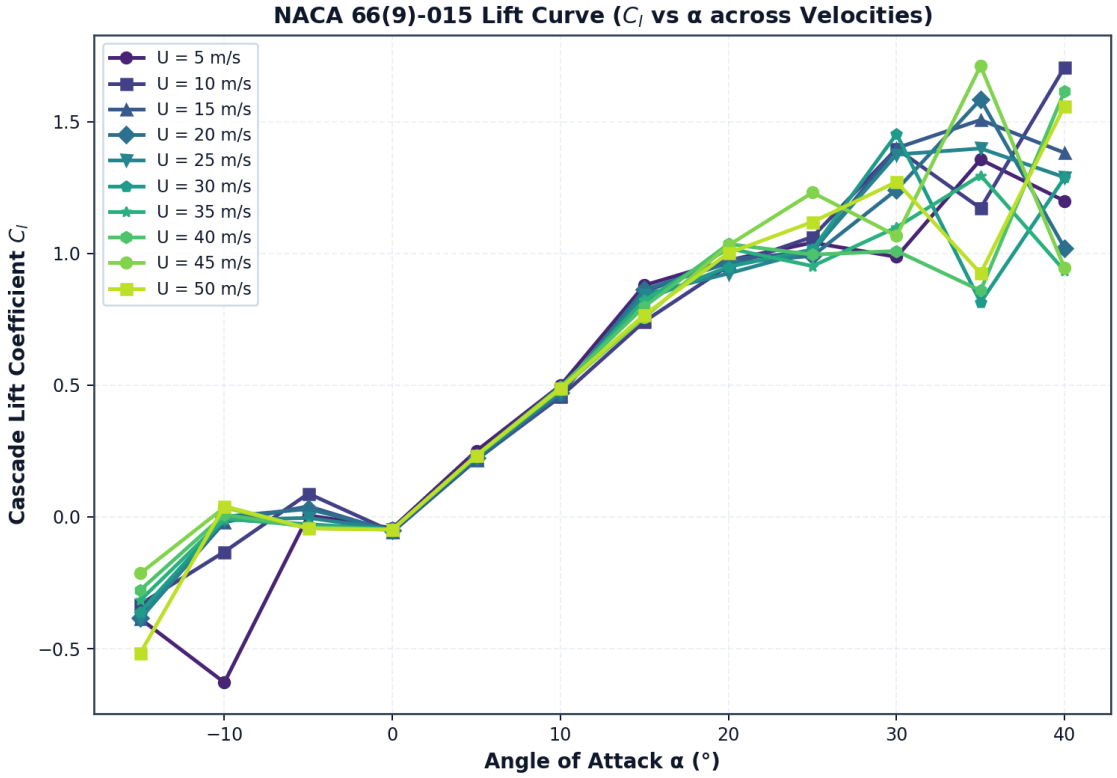


Figure 18: Master Figure: Cascade Cl By Alpha

Significance & Interpretation: Aerodynamic investigation record (cascade_cl_by_alpha.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.

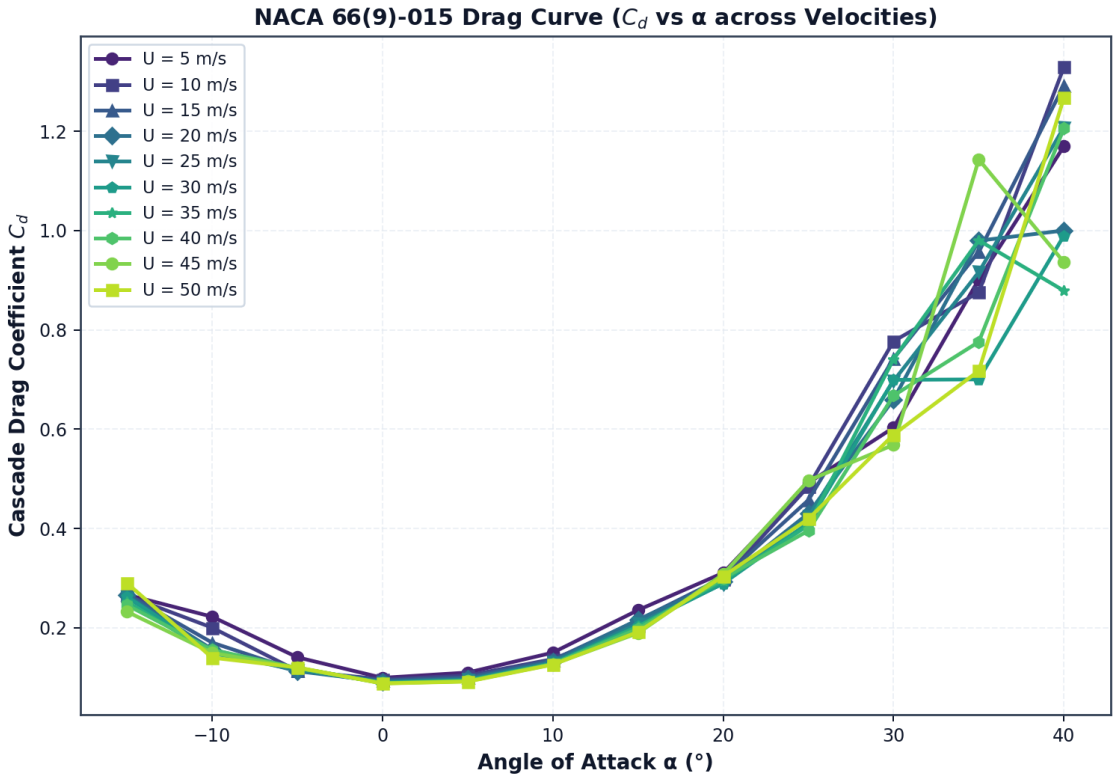


Figure 19: Master Figure: Cascade Cd By Alpha

Significance & Interpretation: Aerodynamic investigation record (cascade_cd_by_alpha.png) for NACA 66(9)-015 cascade operating at U = 15.0 m/s, AoA = +5.0°, Cl = 0.2298, Cd = 0.0973. Evaluated on 38,239-cell Blossom quad mesh.